VE216 Mid-1

Classification of Signals

{continuous ~ {deterministic ~ {energy: $0<E<\infty$
discrete ~ {random ~ {power ~ $0<P<\infty$

even ~ $x_e=\frac{1}{2}(x(t)+x(-t))$ {periodic $x(t)=x(t+T)$
odd ~ $x_o=\frac{1}{2}(x(t)-x(-t))$ {nonperiodic $x[n]=x[n+N]$

conjugate symmetry: $x(-t)=\bar{x}(t)$ $\omega=\frac{2\pi}{T}$ $\Omega=\frac{2\pi}{N}$

Energy & Power

inst. P	$p(t)=\lvert x(t)\rvert^2$	$p[n]=\lvert x[n]\rvert^2$
av. P	$P=\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}\lvert x(t)\rvert^2dt$	$P=\lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N}\lvert x[n]\rvert^2$
E	$E=\int_{-\infty}^{\infty}\lvert x(t)\rvert^2dt$	$E=\sum_{n=-\infty}^{\infty}\lvert x[n]\rvert^2$

For periodic ~, $P=\frac{1}{T}\int_0^T\lvert x(t)\rvert^2dt$; $P=\frac{1}{N}\sum_{n=0}^{N-1}\lvert x[n]\rvert^2$

Root mean square $X_{rms}=\sqrt{\frac{1}{T_2-T_1}\int_{T_1}^{T_2}\lvert x(t)\rvert^2dt}$

Basic Operations

amplitude scaling $y=cx$ addition $y=x_1+x_2$ multiplication $y=x_1x_2$

differentiation $y(t)=\frac{d}{dt}x(t)$ difference $y[n]=x[n]-x[n-1]$

integration $y(t)=\int_{-\infty}^{t}x(\tau)d\tau$ accumulation $y[n]=\sum_{k=-\infty}^{n}x[k]$

time scaling $y(t)=x(\alpha t)$, $y[n]=x[kn]$. $k\in N^*$ | reflection

time shifting $y(t)=x(t-t_0)$, $y[n]=y[n-n_0]$ | $y(t)=x(-t)$ $y[n]=x[-n]$

Elementary Signals

exponential $x(t)=Be^{\alpha t}$ $x[n]=B\alpha^n$ ← real root

Sinusoidal $x(t)=A\cos(\omega t+\varphi)$ $x[n]=A\cos(\Omega n+\varphi)\Rightarrow \Omega N=2\pi m$ ← imaginary root

exponentially damped sinusoidal $e^{j\theta}=\cos\theta+j\sin\theta$ ← complex root

$x(t)=Ae^{-\alpha t}\sin(\omega t+\varphi)$ $(\alpha>0)$ $x[n]=B\alpha^n\sin(\Omega n+\varphi)$ $(0<|\alpha|<1)$

Let $B=Ae^{j\varphi}$ $A\cos(\omega t+\varphi)=\mathrm{Re}\{Be^{j\omega t}\}$; ← 关系

Step function $u(t)=\begin{cases}1, & t>0\\ 0, & t<0\end{cases}$ $u[n]=\begin{cases}1, & n\geq 0\\ 0, & n<0\end{cases}$

Impulse function {$\delta(t)=0$ for $t=0$, $\int_{-\infty}^{+\infty}\delta(t)dt=1$ $\delta[n]=\begin{cases}1, & n=0\\ 0, & n\neq 0\end{cases}$

$\delta(t)=\frac{d}{dt}u(t)$ $u(t)=\int_{-\infty}^{t}\delta(\tau)d\tau$ $\delta[n]=u[n]-u[n-1]$ $u[n]=\sum_{k=-\infty}^{n}\delta[k]$

* sampling $x(t)\delta(t-t_0)=x(t_0)\delta(t-t_0)$ | time-scaling
* shifting $\int_{-\infty}^{\infty}x(t)\delta(t-t_0)dt=x(t_0)$ | $\delta(at)=\frac{1}{|a|}\delta(t)$
* doublet $\delta^{(1)}(t)=\lim_{\Delta\to 0}\frac{1}{\Delta}(\delta(t+\frac{\Delta}{2})-\delta(t-\frac{\Delta}{2}))$

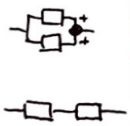
$\int_{-\infty}^{\infty}\delta^{(1)}(t)dt=0$ $\int_{-\infty}^{\infty}f(t)\delta^{(n)}(t-t_0)dt=\frac{d^n}{dt^n}f(t)\big|_{t=t_0}$

Ramp function $r(t)=\begin{cases}t, & t\geq 0\\ 0, & t<0\end{cases}$ $r[n]=\begin{cases}n, & n\geq 0\\ 0, & n<0\end{cases}$

$r(t)=tu(t)$ $u(t)=\frac{d}{dt}r(t)$ $r[n]=nu[n]$ $u[n]=r[n+1]-r[n]$

Interconnection of LTI ~

distributive	$x*h_1+x*h_2=x*(h_1+h_2)$
associative	$(x*h_1)*h_2=x*(h_1*h_2)$
commutative	$h_1*h_2=h_2*h_1$



Convolution Sum

$x[n]=\sum_{k=-\infty}^{\infty}x[k]\delta[n-k]$ $y[n]=\sum_{k=-\infty}^{\infty}x[k]h[n-k]=x[n]*h[n]$

法一 ① 用δ表示x ② 用h表示y

法二 ① 令x=δ求出h ② reflect & shift 确定范围 $x[k]$ $h[n-k]$

③ 求 $W_n[k]=x[k]h[n-k]$ ④ $y[n]=\sum_{k=-\infty}^{\infty}W_n[k]$ * k换为 n-k

Convolution Integral

$x(t)=\int_{-\infty}^{\infty}x(\tau)\delta(t-\tau)d\tau$ $y(t)=\int_{-\infty}^{\infty}x(\tau)h(t-\tau)d\tau=x(t)*h(t)$

① 令 x=δ求出h ② reflect & shift 确定范围 $x(\tau)$ $h(t-\tau)$

③ 求 $W_t(\tau)=x(\tau)h(t-\tau)$ ④ $y(t)=\int_{t=-\infty}^{\infty}W_t(\tau)d\tau$

Properties of Systems (LTI)

memoryless - depend only on current input $h(\tau)=C\delta(\tau)$ $h[k]=C\delta[k]$

causal - depend only on past/present input $h(\tau)=0$ for $\tau<0$ $h[k]=0$ for $k<0$

stable - BI⇒BO $\sum_{k=-\infty}^{\infty}\lvert h[k]\rvert<\infty$ $\int_{-\infty}^{\infty}\lvert h(\tau)\rvert d\tau<\infty$

invertible $h(t)*h^{inv}(t)=\delta(t)$ $h[n]*h^{inv}[n]=\delta[n]$

* ① 令 x=δ 得 y=h ② 令 y=δ 得 x=h^{inv}

Step Response $s(t)=\int_{-\infty}^{t}h(\tau)d\tau$ $s[n]=\sum_{k=-\infty}^{n}h[k]$

$h(t)=\frac{d}{dt}s(t)$ $h[n]=s[n]-s[n-1]$

Differential Equation

$\sum_{k=0}^{N}a_k\frac{d^k}{dt^k}y(t)=\sum_{k=0}^{M}b_k\frac{d^k}{dt^k}x(t)$

① $\sum_{k=0}^{N}a_kr^k=0\Rightarrow y^{(h)}(t)=\sum_{i=1}^{N}C_ie^{r_it}$ 若r_j为p重根 $e^{r_jt}, te^{r_jt}, \dots, t^{p-1}e^{r_jt}$

② 设$y^{(p)}(t)$. 求 ③ $y(t)=y^{(h)}(t)+y^{(p)}(t)$ 代入初始值求.

初始值: $y(t)\big|_{t=0^-}$ $\frac{d}{dt}y(t)\big|_{t=0^-}$ … $\frac{d^{N-1}}{dt^{N-1}}y(t)\big|_{t=0^-}\Rightarrow t=0^+$

Difference Equation

$\sum_{k=0}^{N}a_ky[n-k]=\sum_{k=0}^{M}b_kx[n-k]$

① $\sum_{k=0}^{N}a_kr^{N-k}=0\Rightarrow y^{(h)}[n]=\sum_{i=1}^{N}C_ir_i^n$ 若r_j为p重根 $r_j^n, nr_j^n \dots n^{p-1}r_j^n$

② 设$y^{(p)}[n]$ 求 ③ $y[n]=y^{(h)}[n]+y^{(p)}[n]$ 代入初始值. 求

初始值: $y[-N], y[-N+1] \dots y[-1]\Rightarrow y[0]\, y[1] \dots y[N-1]$

$x(t)$	$y^{(p)}(t)$	$x[n]$	$y^{(p)}[n]$
1	C	1	C
t	C_1t+C_2	n	C_1n+C_2
$e^{-\alpha t}$	$Ce^{-\alpha t}$	α^n	$C\alpha^n$
$\cos(\omega t+\varphi)$	$C_1\cos(\omega t+\varphi)+C_2\sin(\omega t+\varphi)$	"	

connection $\frac{d}{dt}x(t)=\frac{x[n]-x[n-1]}{T_s}$

* nautral response $y^{(n)}(t)[n]\Leftarrow y^{(h)}(0)$ / $y^{(h)}[-1]$
* forced response $y^{(f)}(t)[n]\Leftarrow$ 令初始值为0

Fourier

$x[n]=e^{j\Omega n}$ $y[n]=H(e^{j\Omega})e^{j\Omega n}$ $(H(e^{j\Omega})=\sum_{k=-\infty}^{\infty}h[k]e^{-j\Omega k})$

$x(t)=e^{j\omega t}$ $y(t)=H(j\omega)e^{j\omega t}$ $(H(j\omega)=\int_{-\infty}^{\infty}h(\tau)e^{-j\omega\tau}d\tau)$

For DTFS $\hat{x}[n]=\sum_{k=0}^{N-1}A[k]e^{jk\Omega_0 n}$ $MSE=\frac{1}{N}\sum_{n=0}^{N-1}\lvert x[n]-\hat{x}[n]\rvert^2$ mean-square error

For FS $\hat{x}(t)=\sum_{k=-\infty}^{\infty}A[k]e^{jk\omega_0 t}$ $MSE=\frac{1}{T}\int_0^T\lvert x(t)-\hat{x}(t)\rvert^2dt$

function of frequency

	periodic	nonperiodic
continuous	FS	FT
discrete	DTFS	DTFT

$C=a+jb$

$C=|C|e^{j\arg\{C\}}$

where $|C|=\sqrt{a^2+b^2}$ $\arg\{C\}=\arctan(\frac{b}{a})$

$\cos^2\theta=\frac{1+\cos 2\theta}{2}$ $\sin^2\theta=\frac{1-\cos 2\theta}{2}$

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